Business Question and Visualization Report

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

Business Questions and Visualisation

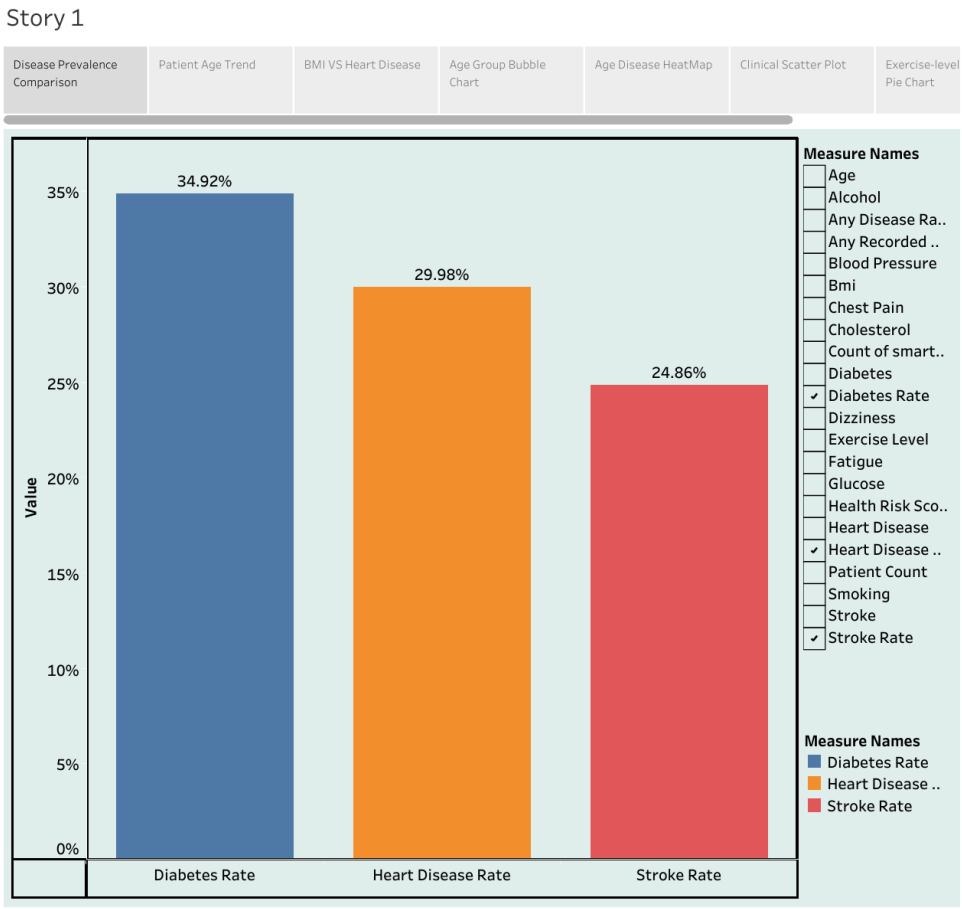
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake / Tableau workspace. Well-framed business questions help identify the required metrics, determine the appropriate analytical views, select relevant data, and create visualizations that provide useful business insights.

Question 1 (Disease Prevalence Comparison): What is the baseline prevalence rate across the three major chronic conditions (Diabetes Rate, Heart Disease Rate, and Stroke Rate) in the patient population?

Code:

```
SELECT
    ROUND(AVG(Diabetes_Rate) * 100, 2) AS Diabetes_Rate,
    ROUND(AVG(Heart_Disease_Rate) * 100, 2) AS Heart_Disease_Rate,
    ROUND(AVG(Stroke_Rate) * 100, 2) AS Stroke_Rate
FROM HEALTH360_PATIENT_DB.CLINICAL_SUMMARY;
```

Screenshot of visualisation:

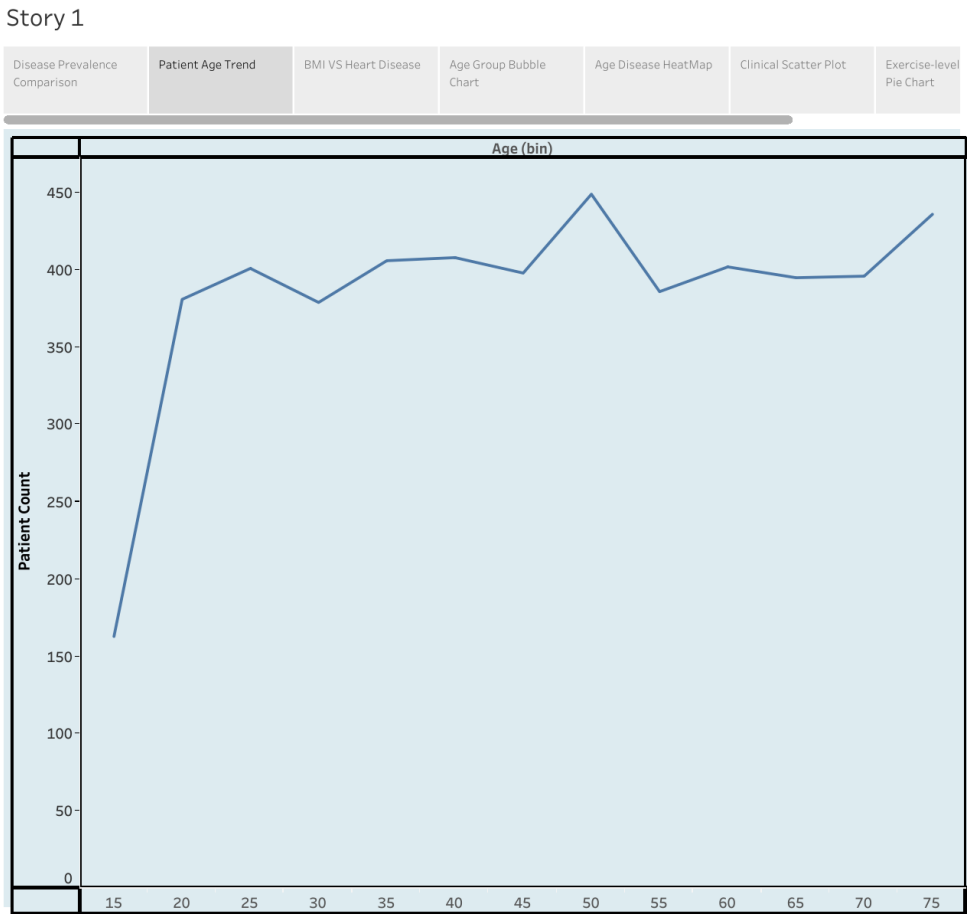


Question 2 (Patient Age Trend): How is the patient count distributed across age bins (15 to 75+ years), and what are the peak demographic age cohorts?

Code:

```
SELECT
    Age_Bin,
    COUNT(Patient_ID) AS Patient_Count
FROM HEALTH360_PATIENT_DB.DEMOGRAPHIC_RECORDS
GROUP BY Age_Bin
ORDER BY Age_Bin ASC;
```

Screenshot of visualisation:

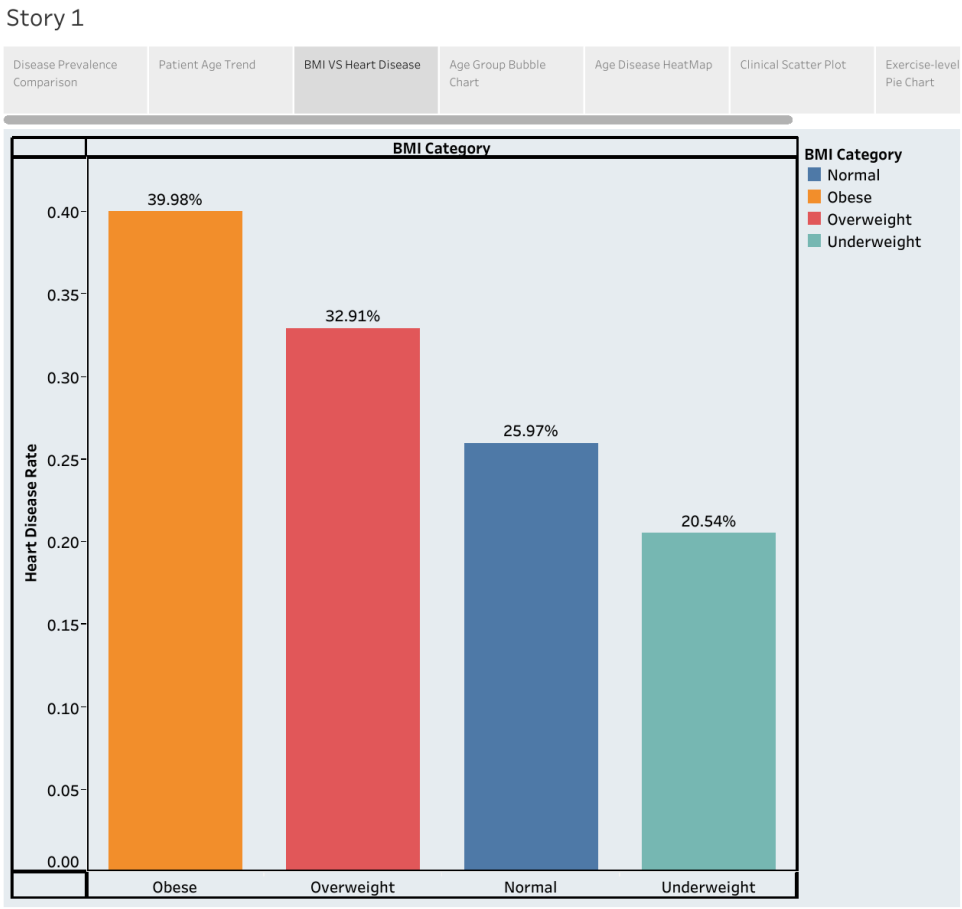


Question 3 (BMI VS Heart Disease): How does the Heart Disease prevalence rate vary across BMI categories (Obese, Overweight, Normal, Underweight)?

Code:

```
SELECT
    BMI_Category,
    ROUND(AVG(Heart_Disease_Flag), 4) AS Heart_Disease_Rate
FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS
GROUP BY BMI_Category
ORDER BY Heart_Disease_Rate DESC;
```

Screenshot of visualisation:



Question 4 (Age Group Bubble Chart): What is the overall disease burden rate across broad age group ranges (18-29, 30-44, 45-59, 60-74, 75-79)?

Code:

```
SELECT
    Age_Group_Range,
    ROUND(AVG(Any_Disease_Rate), 3) AS Any_Disease_Rate,
    COUNT(Patient_ID) AS Total_Patients
FROM HEALTH360_PATIENT_DB.DEMOGRAPHIC_RECORDS
GROUP BY Age_Group_Range;
```

Screenshot of visualisation:



Question 5 (Age Disease HeatMap): What is the intensity matrix of Diabetes, Heart Disease, and Stroke rates across demographic life stages (Adult, Middle Age, Old, Senior, Teenager, Young Adult)?

Code:

```
SELECT
    Age_Group,
    ROUND(AVG(Diabetes_Flag) * 100, 1) AS Diabetes_Rate_Pct,
    ROUND(AVG(Heart_Disease_Flag) * 100, 1) AS Heart_Disease_Rate_Pct,
    ROUND(AVG(Stroke_Flag) * 100, 1) AS Stroke_Rate_Pct
FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS
GROUP BY Age_Group;
```

Screenshot of visualisation:

Story 1

Patient Age Trend | BMI VS Heart Disease | Age Group Bubble Chart | Age Disease HeatMap | Clinical Scatter Plot | Exercise-level Pie Chart | Smoking Status

Age Group	Diabetes Rate	Heart Disease Rate	Stroke Rate
Adult	0.0%	0.0%	0.0%
Middle Age	21.8%	11.7%	0.0%
Old	100.0%	100.0%	100.0%
Senior	100.0%	79.7%	60.0%
Teenager	0.0%	0.0%	0.0%
Young Adult	0.0%	0.0%	0.0%

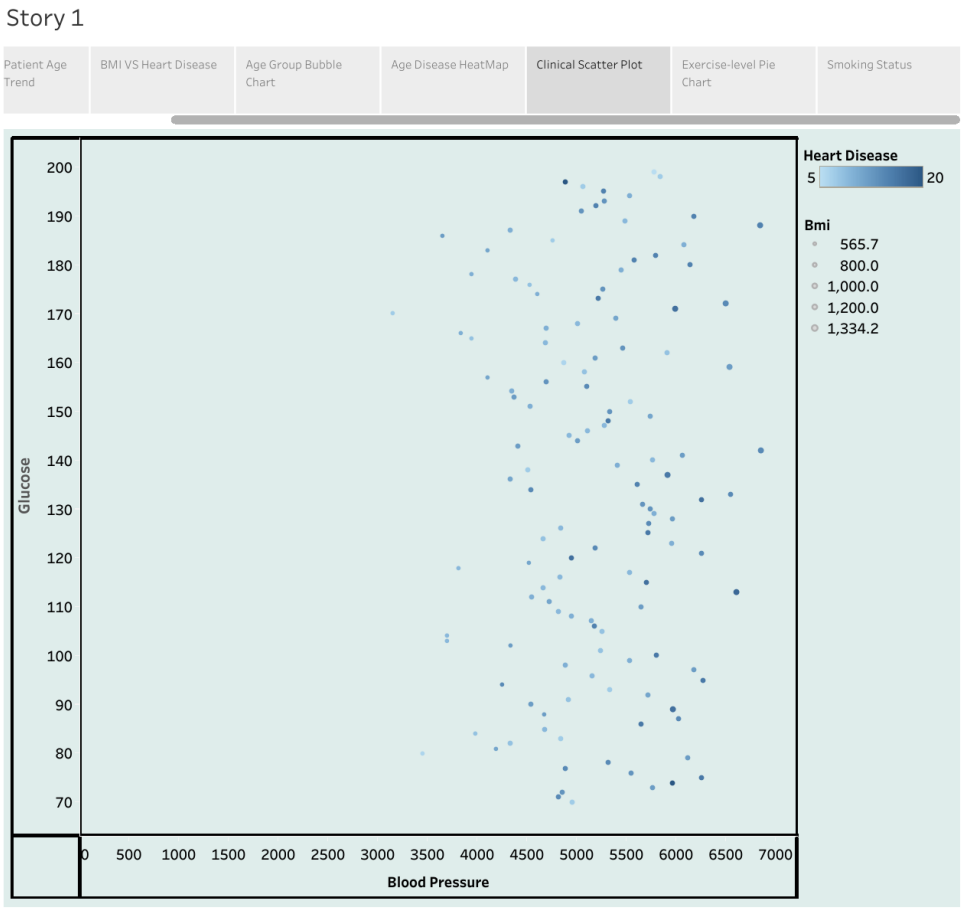
Measure Values
0.0% 100.0%

Question 6 (Clinical Scatter Plot): What is the clinical correlation between Systolic Blood Pressure and Glucose levels when layered with Heart Disease status and BMI dimensions?

Code:

```
SELECT
    Patient_ID, Blood_Pressure, Glucose, Heart_Disease_Status, BMI
FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS
WHERE Blood_Pressure IS NOT NULL AND Glucose IS NOT NULL;
```

Screenshot of visualisation:

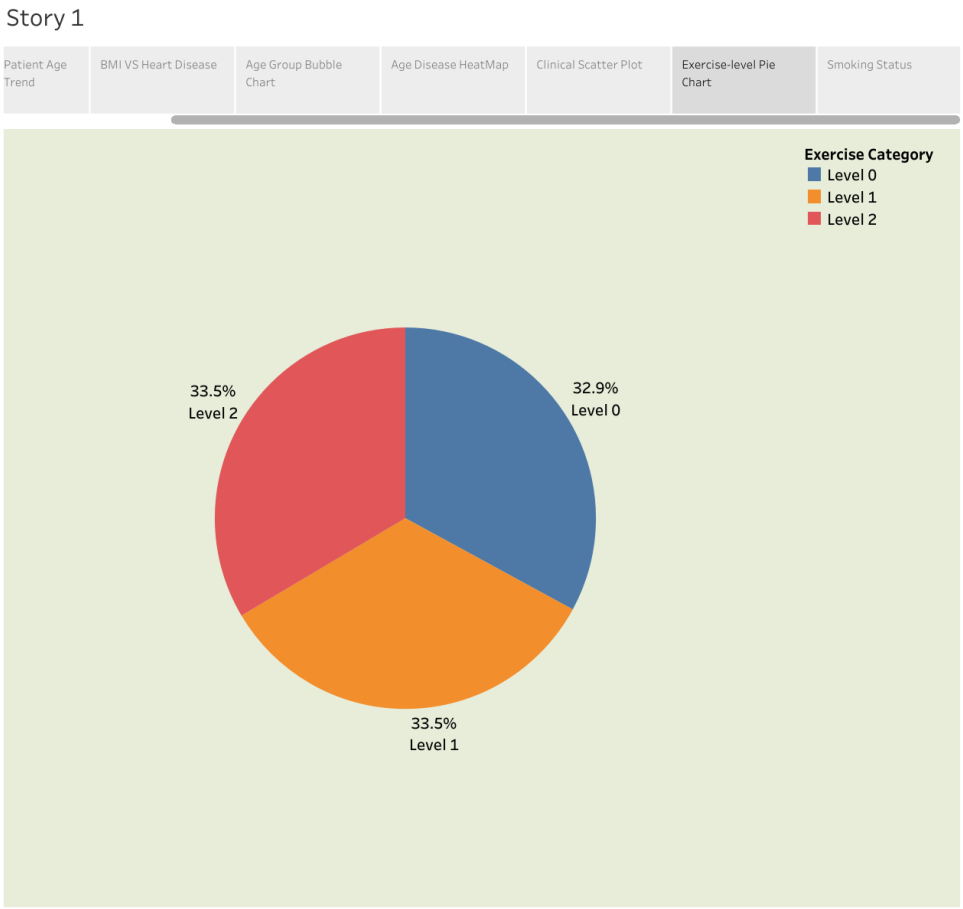


Question 7 (Exercise-level Pie Chart): What is the percentage distribution of patients across physical activity categories (Level 0, Level 1, Level 2)?

Code:

```
SELECT
    Exercise_Category,
    COUNT(Patient_ID) AS Count_Patients,
    ROUND(COUNT(Patient_ID) * 100.0 / SUM(COUNT(Patient_ID)) OVER(), 1) AS Pct_Distribution
FROM HEALTH360_PATIENT_DB.LIFESTYLE_RECORDS
GROUP BY Exercise_Category;
```

Screenshot of visualisation:



Question 8 (Smoking Status Analysis): Does Smoking Status (Yes vs. No) impact the Heart Disease Rate among patients?

Code:

```
SELECT
    Smoking_Status,
    COUNT(Patient_ID) AS Total_Patients,
    ROUND(AVG(Heart_Disease_Flag) * 100, 1) AS Heart_Disease_Rate_Pct
FROM HEALTH360_PATIENT_DB.LIFESTYLE_RECORDS
GROUP BY Smoking_Status;
```

Screenshot of visualisation:

Story 1

Patient Age Trend | BMI VS Heart Disease | Age Group Bubble Chart | Age Disease HeatMap | Clinical Scatter Plot | Exercise-level Pie Chart | Smoking Status

